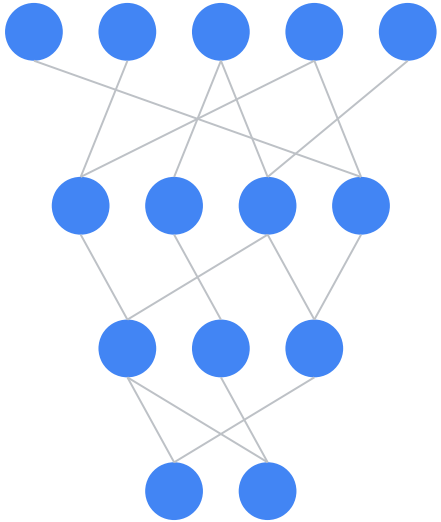
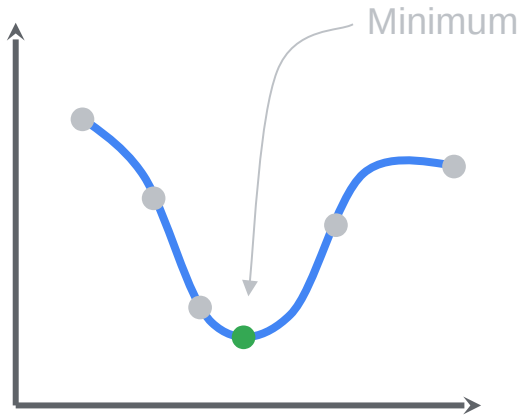
Preview of TinyML Applications

Acoustic Sensors
Ultrasonic, **Microphones**, Geophones, Vibrometers

Image Sensors
Thermal, **Image**

Motion Sensors
Gyroscope, Radar, **Accelerometer**





TinyML Application Areas



Home



Office



Industry

TinyML Application Areas



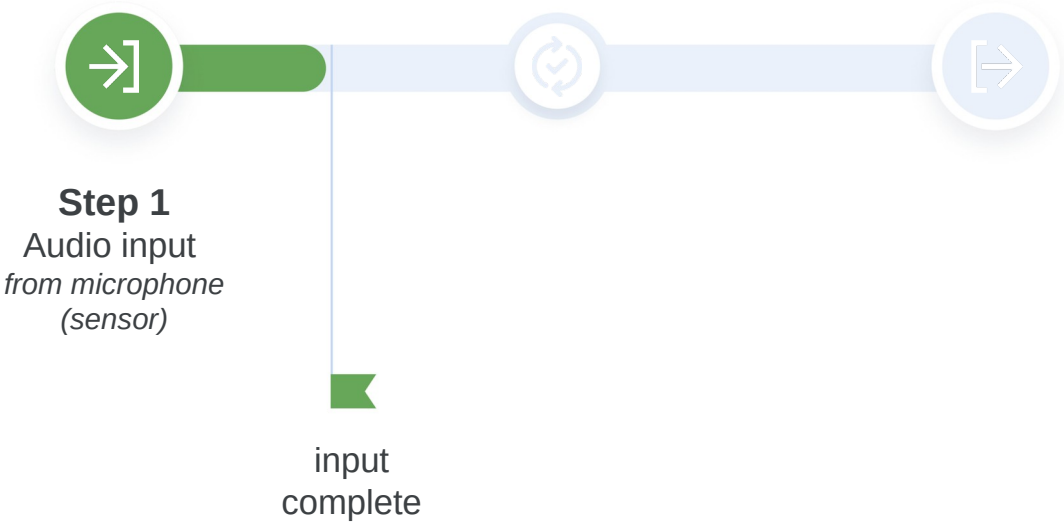
Home

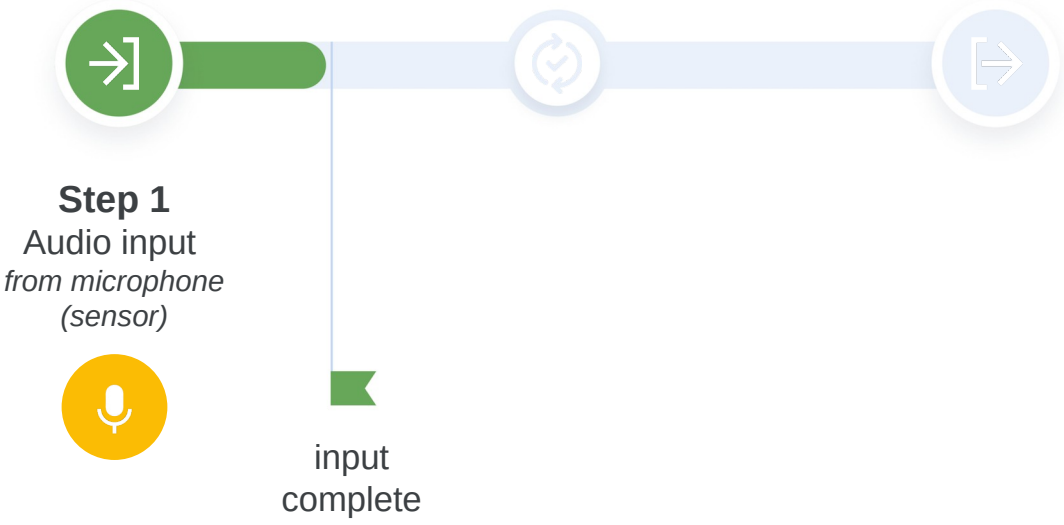
Office

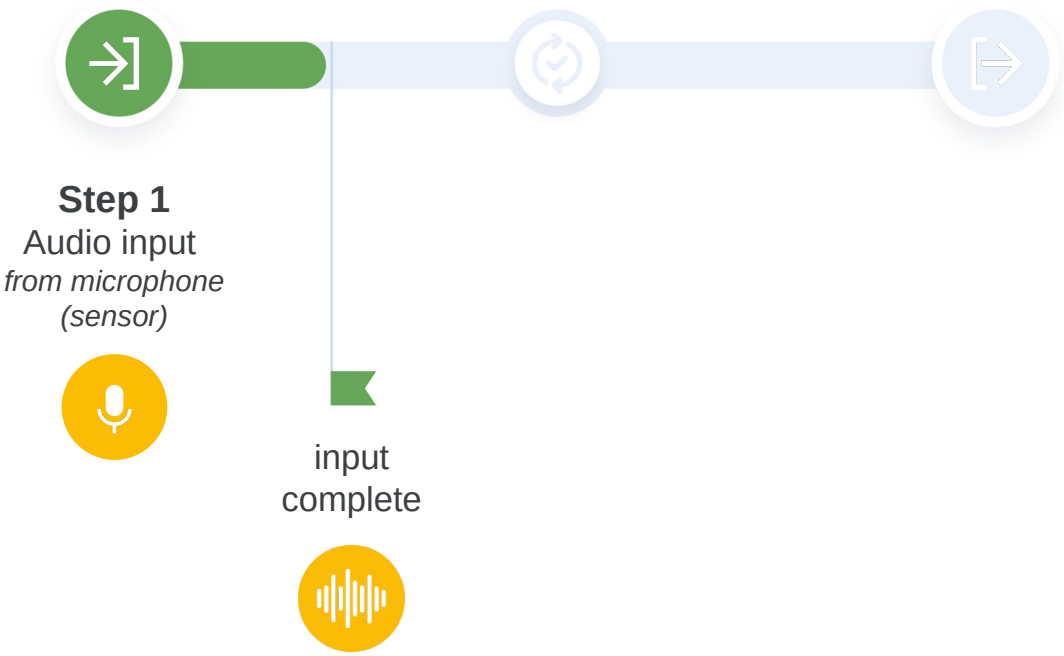
Industry

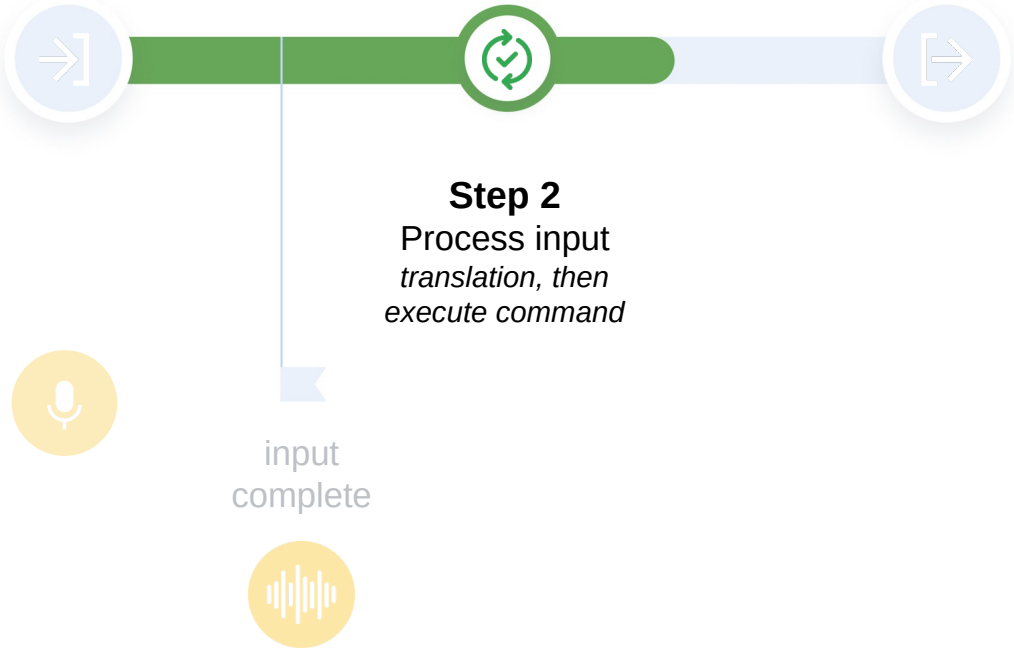
Keyword Spotting

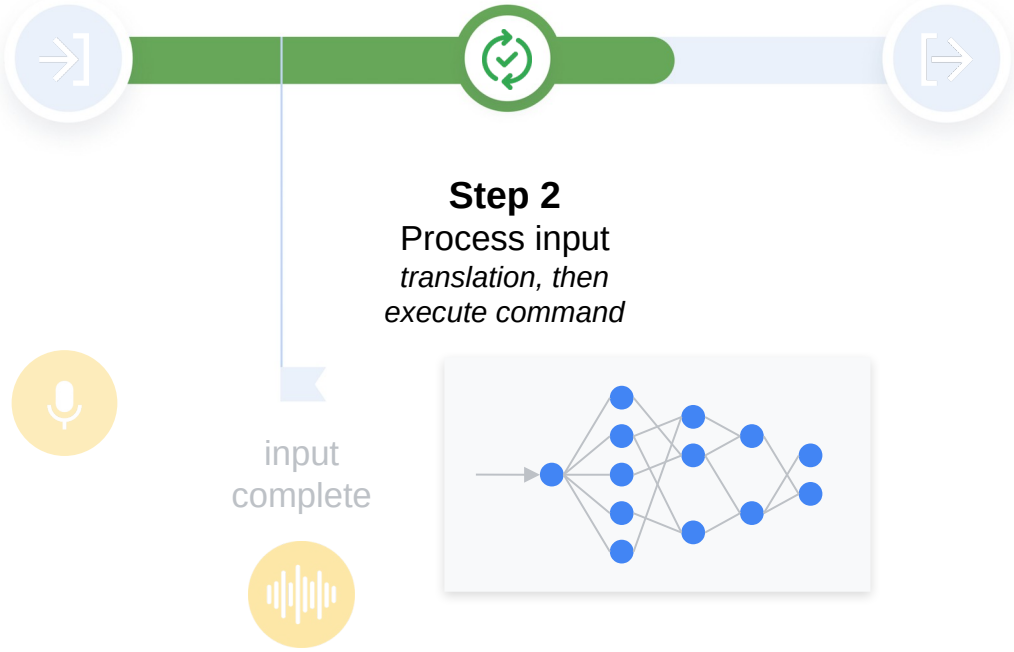


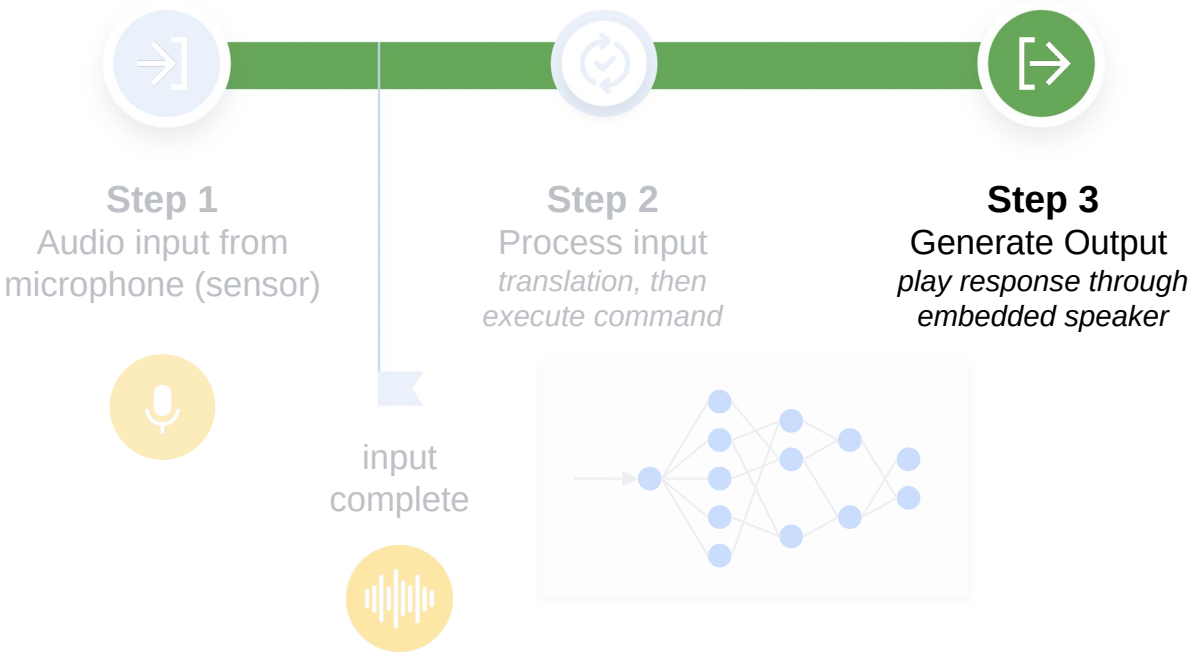

Step 1
Audio input
from microphone
(sensor)
input
complete

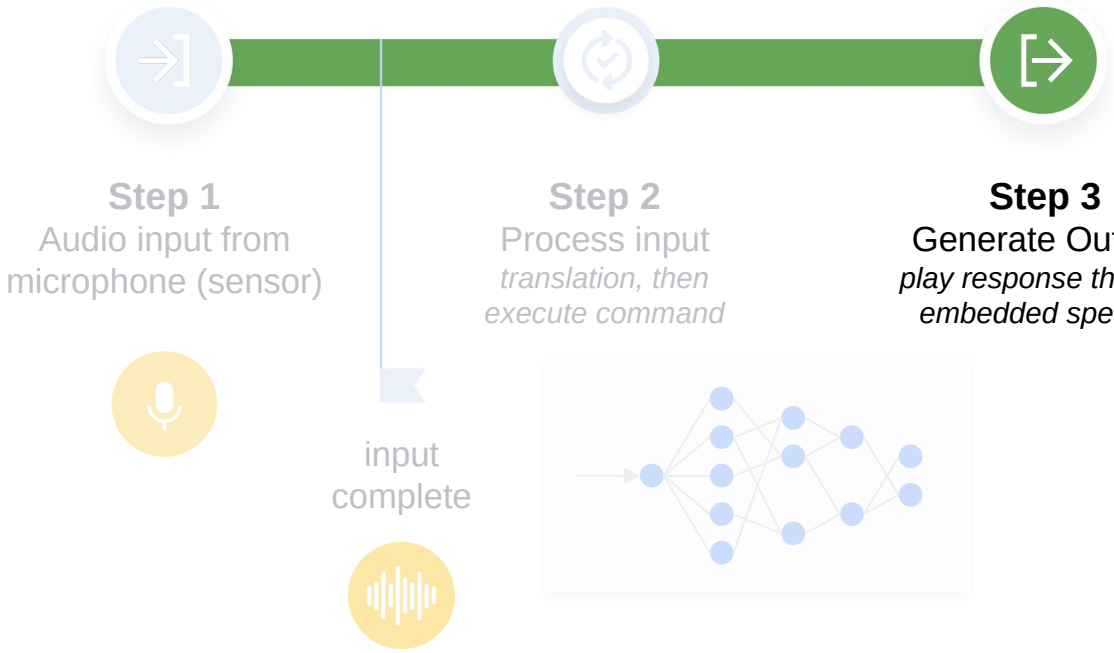

Step 1
Audio input
from microphone
(sensor)
input
complete


Step 1
Audio input
from microphone
(sensor)
input
complete


Step 2
Process input
translation, then
execute command
input
complete

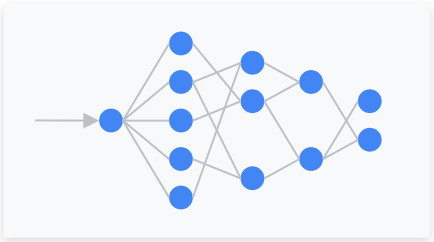

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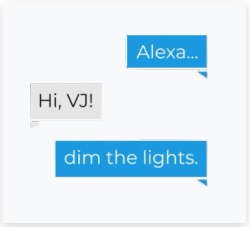

Step 1
Audio input from microphone (sensor)
input complete
Step 2
Process input
translation, then execute command
Step 3
Generate Output
play response through embedded speaker


Alexa...
Hi, VJ!
dim the lights.
Step 1
Audio input from microphone (sensor)
input complete
Step 2
Process input
translation, then execute command
Step 3
Generate Output
play response through embedded speaker








Alexa...
Hi, VJ!
dim the lights.



Questions

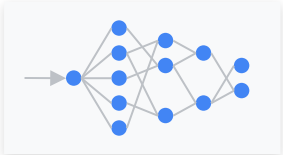


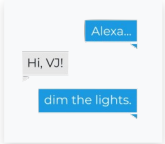
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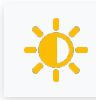
Questions







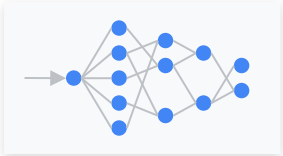
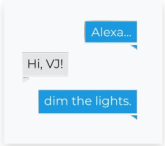




- How do we **capture** the data to feed into the neural network?
- How do you **design** the neural network to take in the speech signal?

Questions



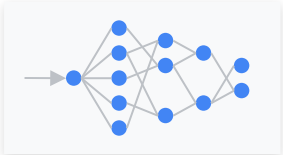
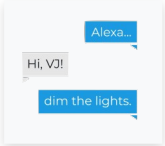



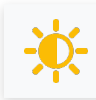


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Questions



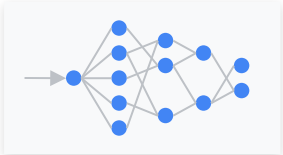
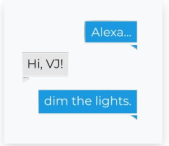



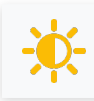


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- What **dataset** does the neural network need to be trained?
- How do we **pre-process** the data for neural network inference?

Questions



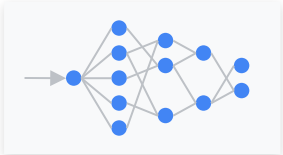


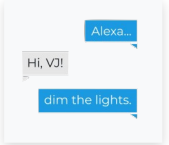
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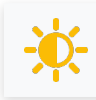
Questions







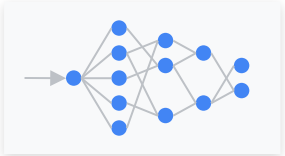
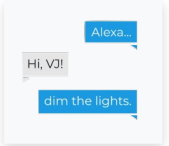




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Questions





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TinyML Application Areas

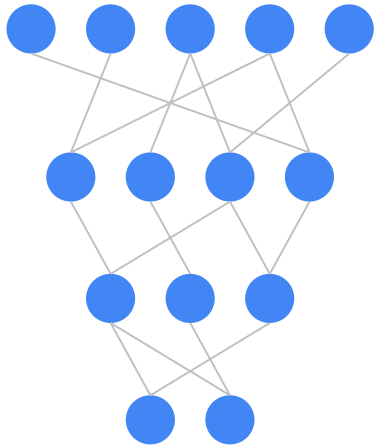
Home

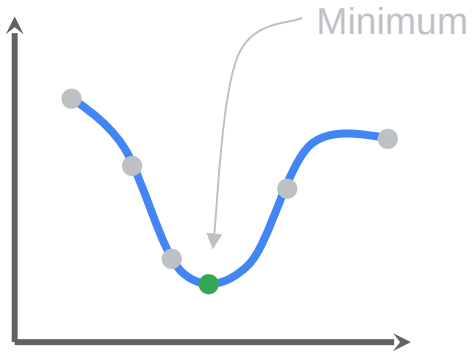


Office

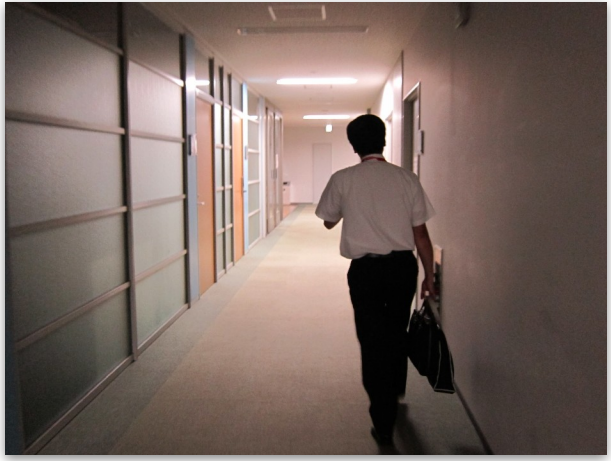
Industry

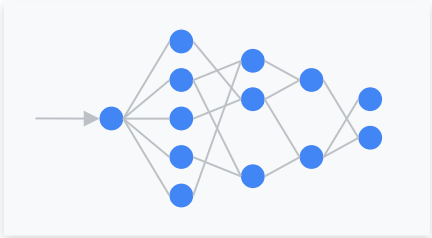
Image Sensors
Thermal, **Image**

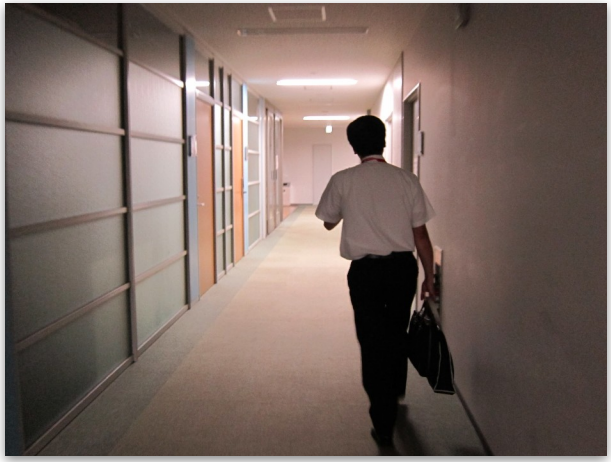


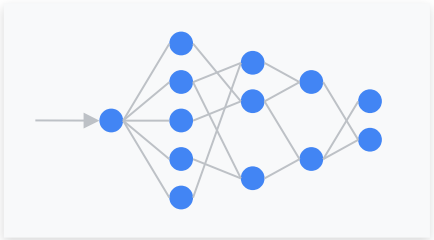


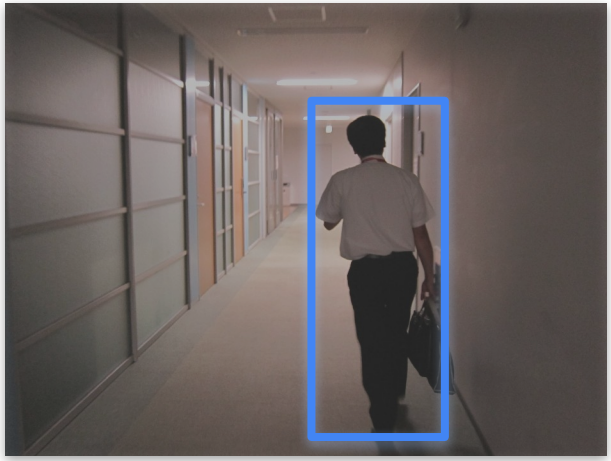






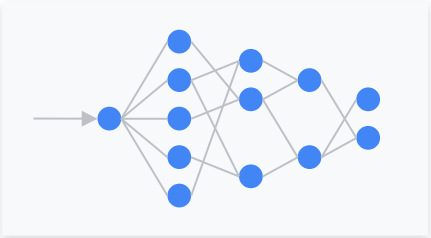






ML Model Evolution

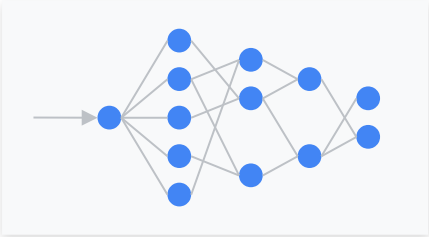
- **MobileNet** (2015)
  - ***MobileNetv1***
    - 70.6% accuracy
    - 16.9MB in size



Source: S. Bianco, R. Cadene, L. Celona, and P. Napoletano, “Benchmark analysis of representative deep neural network architectures,” *IEEE Access*, vol. 6, pp. 64 270–64 277, 2018

ML Model Evolution

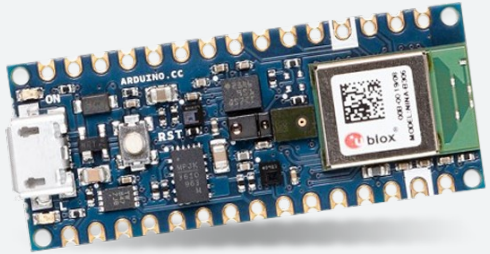
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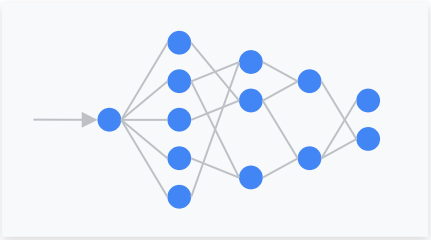
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Problem

Our board (in your kit for Course 3) only has **256KB** of RAM (memory) yet ***MobileNetv1*** needs **16.9MB**!



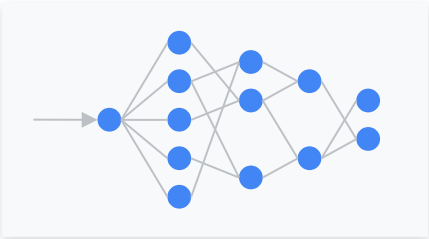
How do we engineer a **TinyML** vision network?



Think:

- Compute **operations**

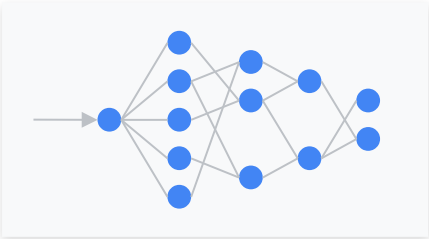
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Think:

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- Operator **numerics**

How do we engineer a **TinyML** vision network?

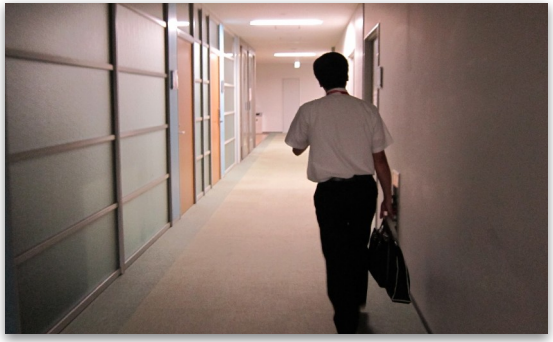


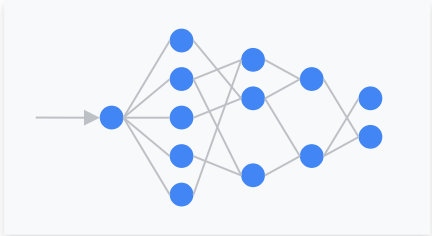
Think:

- Compute **operations**
- Operator **numerics**
- **Compression** methods (e.g., pruning, quantization)

What is the **end-to-end** workflow?

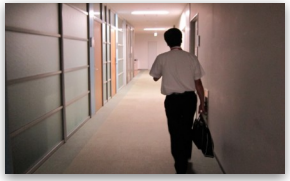






Questions



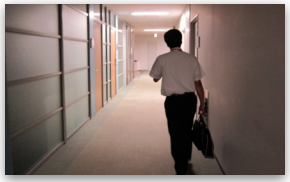




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- How do you make sure there is no **bias** in the dataset?
- How do you **deploy** this on the microcontroller?

Questions



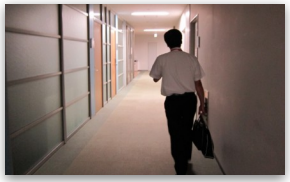




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- *How do we ensure that the neural network is **resilient**?*

Questions







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- *How do we ensure that the neural network is **resilient**?*
- *How do we get the neural network to **train faster**?*

TinyML Application Areas

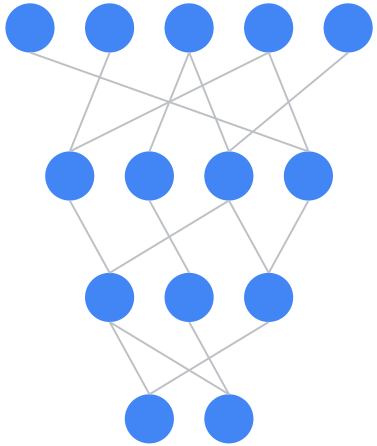
Home

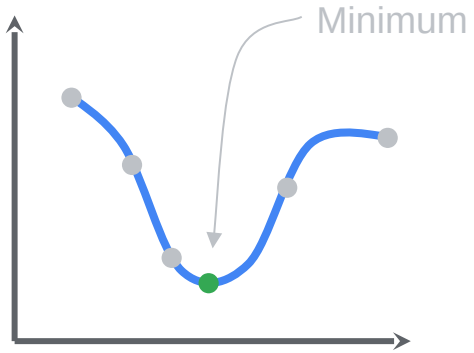
Office

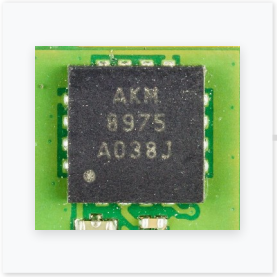


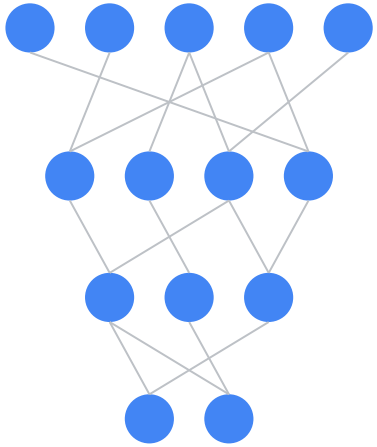
Industry

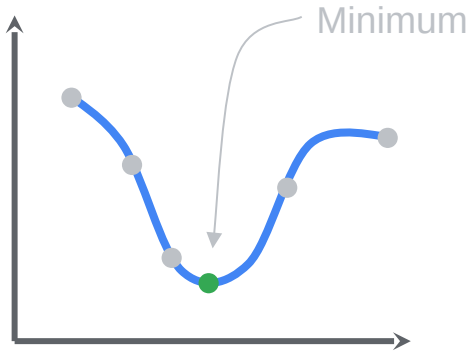
Motion Sensors
Gyroscope, Radar,
Accelerometer



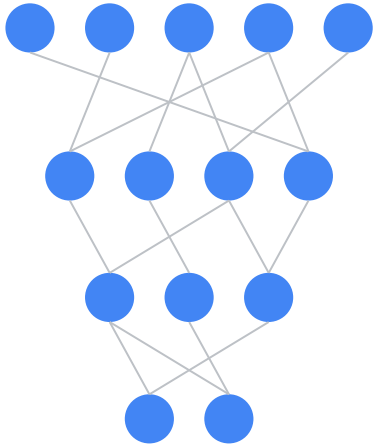


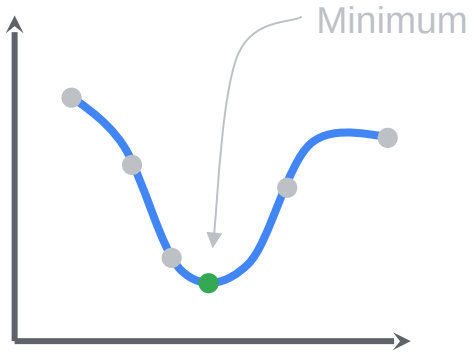

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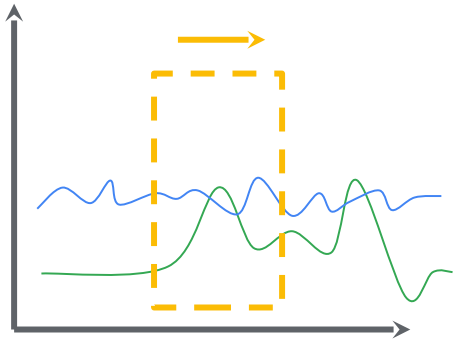


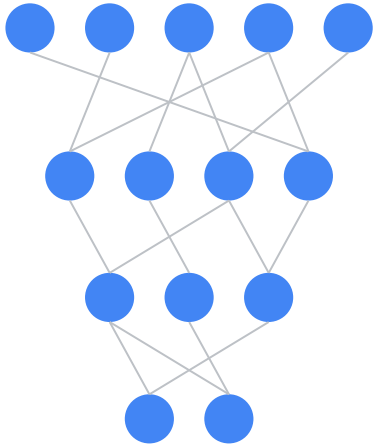

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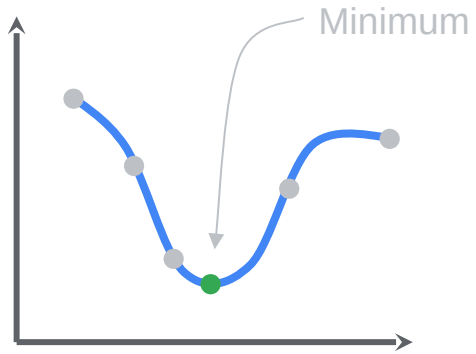


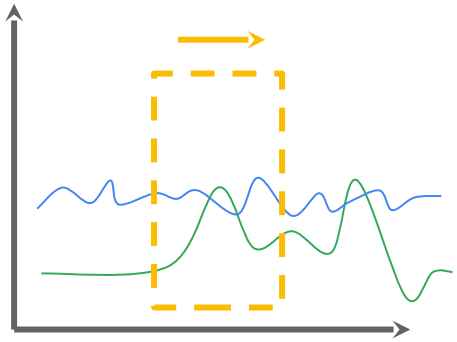


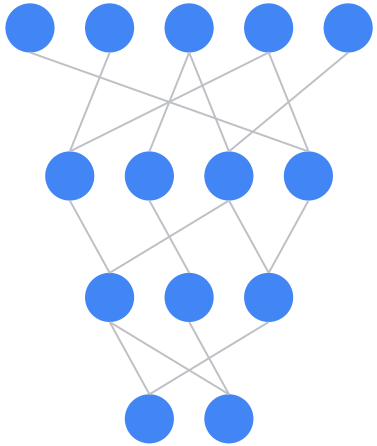

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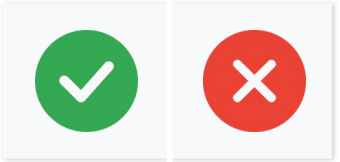




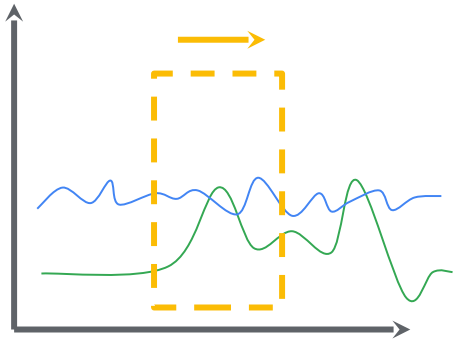

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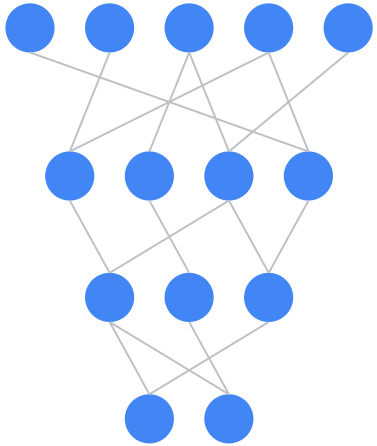


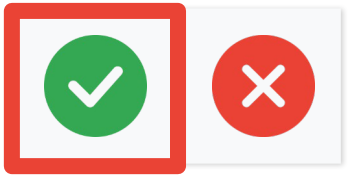




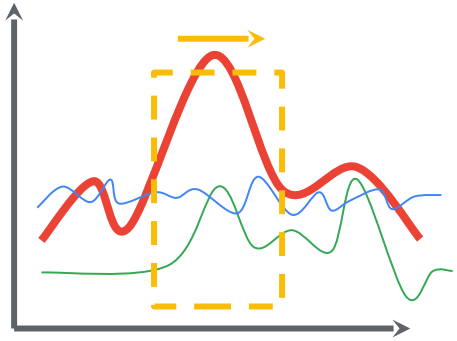
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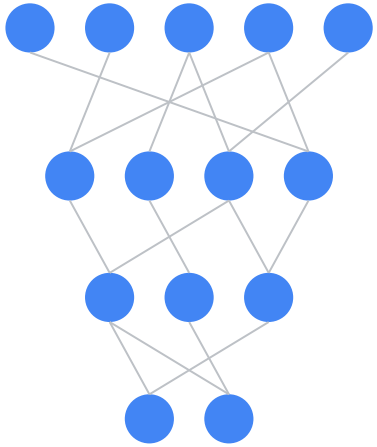




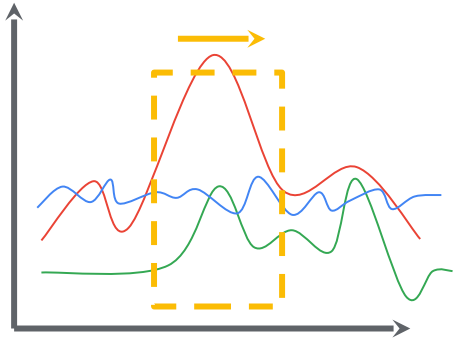


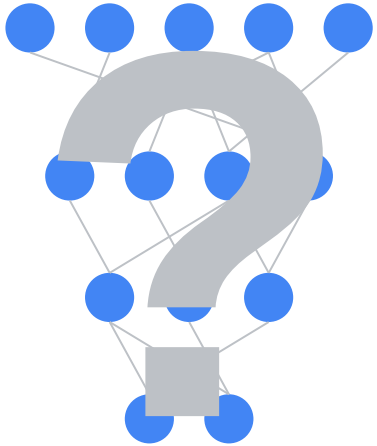
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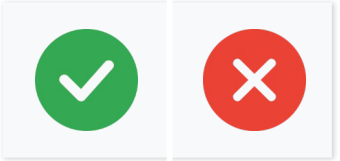






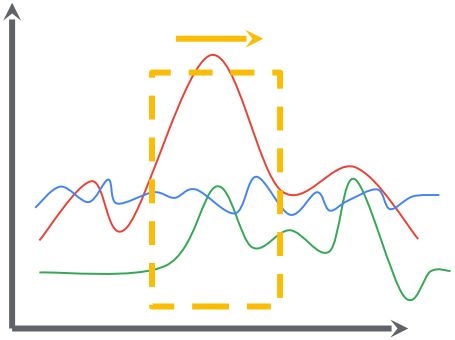


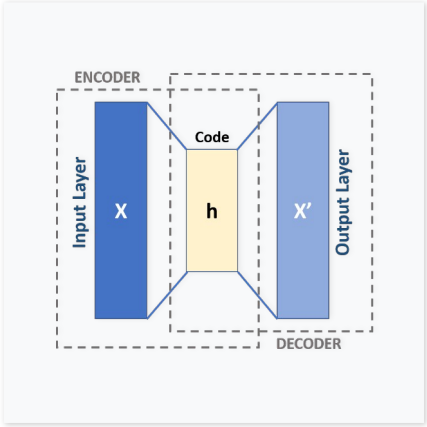


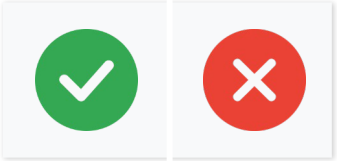


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Autoencoder

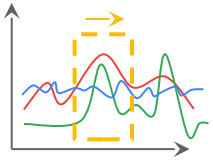


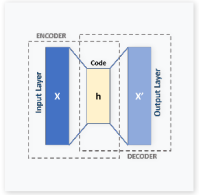


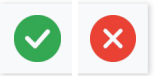


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Questions







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- *How do we **capture** the **time-series** data to feed into the neural network?*
- How do we **pre-process** the data for neural network inference?
- *How do you **design** the **autoencoder** neural network?*
- What **dataset** does the neural network need to be trained?
- How do you **post-process** the neural network output?
- How do you make sure there is no **bias** in the dataset?
- How do you **deploy** this on the microcontroller?

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